Table 43.1: Elementary Particles and Their Properties.

Particles/ Anti-particles	**Symbol**	**Charge** e	**Mass** MeV	**Lifetime** s	**Spin** $\hbar$	**Baryon Charge**	**Strangeness**	**Isospin**	**Isospin Projection**	**Dominant Decay Mode**
Photon	γ	0	0	Stable	1	0	–	–	–	Stable
Leptons										
Neutrinos										
Electron Neutrino	ν_e	0	0	Stable	1/2	0	0	1/2	+1/2	Stable
Electron Antineutrino	$\tilde{\nu}_e$	0	0	Stable	1/2	0	0	1/2	−1/2	Stable
Muon Neutrino	ν_μ	0	0	Stable	1/2	0	0	1/2	+1/2	Stable
Muon Antineutrino	$\tilde{\nu}_\mu$	0	0	Stable	1/2	0	0	1/2	−1/2	Stable
Electrons										
Electron	e^-	−1	0.511	Stable	1/2	0	0	1/2	−1/2	Stable
Positron	e^+	+1	0.511	Stable	1/2	0	0	1/2	+1/2	
Muons										
Muon$^-$	μ^-	−1	105.7	2.2×10^{-6}	1/2	0	0	1/2	−1/2	$\mu^- \to e^- + \tilde{\nu}_e + \nu_\mu$
Muon$^+$	μ^+	+1	105.7	2.2×10^{-6}	1/2	0	0	1/2	+1/2	$\mu^+ \to e^+ + \nu_e + \tilde{\nu}_\mu$
Mesons										
Pions										
π^+ Meson	π^+	+1	139.6	2.6×10^{-8}	0	0	0	1	+1	$\pi^+ \to \mu^+ + \nu_\mu$
π^- Meson	π^-	−1	139.6	2.6×10^{-8}	0	0	0	1	−1	$\pi^- \to \mu^- + \tilde{\nu}_\mu$
π^0 Meson	π^0	0	135.0	8.4×10^{-17}	0	0	0	1	0	$\pi^0 \to \gamma + \gamma$
Kaons										
Kaon$^+$	K^+	+1	493.7	1.24×10^{-8}	0	0	+1	1/2	+1/2	$K^+ \to \mu^+ + \nu_\mu$
Kaon$^-$	K^-	−1	493.7	1.24×10^{-8}	0	0	−1	1/2	−1/2	$K^- \to \mu^- + \tilde{\nu}_\mu$

Continued on next page

Particles/ Anti-particles	**Symbol**	**Charge** e	**Mass** MeV	**Lifetime** s	**Spin** $\hbar$	**Baryon Charge**	**Strangeness**	**Isospin**	**Isospin Projection**	**Dominant Decay Mode**
K^0 Meson	K^0	0	497.6	$K_0^S : 8.6 \times 10^{-11}$	0	0	+1	1/2	−1/2	$K_S^0 \to \pi^+ + \pi^-$ $K_S^0 \to \pi^0 + \pi^0$
anti-K^0 Meson	$\tilde{K}^0$	0	497.6	$K_0^L : 5.38 \times 10^{-8}$	0	0	−1	1/2	+1/2	$K_L^0 \to \pi^\pm + e^\mp + \nu_e$ $K_L^0 \to \pi^\pm + \mu^\mp + \nu_\mu$ $K_L^0 \to \pi^+ + \pi^- + \pi^0$
η^0 Meson	η^0	0	547.9	5.0×10^{-19}	0	0	0	0	0	$\eta^0 \to \gamma + \gamma$ $\eta^0 \to \pi^+ + \pi^- + \pi^0$ $\eta^0 \to 3\pi^0$
Baryons										
Nucleons										
Proton	p	+1	938.3	Stable	1/2	+1	0	1/2	+1/2	
Antiproton	$\tilde{p}$	−1	938.3	Stable	1/2	−1	0	1/2	−1/2	
Neutron	n	0	939.6	880 s	1/2	+1	0	1/2	−1/2	$n \to p + e^- + \tilde{\nu}_e$
Antineutron	$\tilde{n}$	0	939.6	880 s	1/2	−1	0	1/2	+1/2	$\tilde{n} \to \tilde{p} + e^+ + \nu_e$
Hyperons										
Λ^0 Hyperon	Λ^0	0	1115.7	2.6×10^{-10}	1/2	+1	−1	0	0	$\Lambda^0 \to p + \pi^-$
Anti-Λ^0 Hyperon	$\tilde{\Lambda}^0$	0	1115.7	2.6×10^{-10}	1/2	−1	+1	0	0	$\tilde{\Lambda}^0 \to \tilde{p} + \pi^+$
Σ^+ Hyperon	Σ^+	+1	1189.4	8×10^{-11}	1/2	+1	−1	1	+1	$\Sigma^+ \to p + \pi^0$, $\Sigma^+ \to n + \pi^+$
Anti-Σ^+Hyperon	$\tilde{\Sigma}^+$	−1	1189.4	8×10^{-11}	1/2	−1	+1	1	−1	$\tilde{\Sigma}^+ \to \tilde{p} + \pi^0$, $\tilde{\Sigma}^+ \to \tilde{n} + \pi^+$
Σ^- Hyperon	Σ^-	−1	1197.4	1.5×10^{-10}	1/2	+1	−1	1	−1	$\Sigma^- \to n + \pi^-$
Anti-Σ^+ Hyperon	$\tilde{\Sigma}^+$	+1	1197.4	1.5×10^{-10}	1/2	−1	+1	1	+1	$\tilde{\Sigma}^- \to \tilde{n} + \pi^+$
Σ^0 Hyperon	Σ^0	0	1192.6	7.4×10^{-20}	1/2	+1	−1	1	0	$\Sigma^0 \to \Lambda^0 + \gamma$
Anti-Σ^0 Hyperon	$\tilde{\Sigma}^0$	0	1192.6	7.4×10^{-20}	1/2	−1	+1	1	0	$\tilde{\Sigma}^0 \to \tilde{\Lambda}^0 + \gamma$
Ξ^- Hyperon	Ξ^-	−1	1321.7	1.6×10^{-10}	1/2	+1	−2	1/2	−1/2	$\Xi^- \to \Lambda^0 + \pi^-$

Continued on next page

Particles/ Anti-particles	Symbol	Charge e	Mass MeV	Lifetime s	Spin $\hbar$	Baryon Charge	Strangeness	Isospin	Isospin Projection	Dominant Decay Mode
Anti-Ξ^+ Hyperon	$\tilde{\Xi}^+$	$+1$	1321.7	1.6×10^{-10}	$1/2$	-1	$+2$	$1/2$	$+1/2$	$\tilde{\Xi}^- \rightarrow \tilde{\Lambda}^0 + \pi^+$
Ξ^0 Hyperon	Ξ^0	0	1314.9	2.9×10^{-10}	$1/2$	$+1$	-2	$1/2$	$+1/2$	$\Xi^0 \rightarrow \Lambda^0 + \pi^0$
Anti-Ξ^0 Hyperon	$\tilde{\Xi}^0$	0	1314.9	2.9×10^{-10}	$1/2$	-1	$+2$	$1/2$	$-1/2$	$\tilde{\Xi}^0 \rightarrow \tilde{\Lambda^0} + \pi^0$
Ω^- Hyperon	Ω^-	-1	1672.4	8.2×10^{-11}	$3/2$	$+1$	-3	0	0	$\Omega^- \rightarrow \Lambda^0 + K^-$ $\Omega^- \rightarrow \Xi^0 + \pi^-$ $\Omega^- \rightarrow \Xi^- + \pi^0$
Anti-Ω^+ Hyperon	$\tilde{\Omega}^+$	$+1$	1672.4	8.2×10^{-11}	$3/2$	-1	$+3$	0	0	$\tilde{\Omega}^+ \rightarrow \tilde{\Lambda}^0 + K^+$ $\tilde{\Omega}^+ \rightarrow \tilde{\Xi}^0 + \pi^+$ $\tilde{\Omega}^- \rightarrow \tilde{\Xi}^+ + \pi^0$